

Pritesh Rai

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EDUCATION

National Institute Of Technology Karnataka

Master Of Science in Physics

Surathkal, Karnataka

Pursuing

U.P. (Autonomous) College

Bachelor of Science (Physics, Chemistry, Mathematics)

Varanasi, U.P.

Aug. 2021 – May 2024

PROJECT

Double Perovskites Solar Cell

National Institute Of Technology Karnataka

Aug. 2025 – Present

Surathkal, Karnataka

- Synthesis of Double Perovskite materials and controlled doping
- Working to reduce band gap for photovoltaic application
- DFT simulations of material properties

Tin (Sn) based Perovskite Solar Cell

Indian Institute Of Technology (BHU)

May 2025 – July 2025

Varanasi, U.P.

- Synthesis of Tin (Sn) and Lead (Pb)-based perovskites
- Characterization: **XRD, UV-Visible, Conductivity measurement, IV characteristics, Impedance analysis**

DFT Calculations of Nb₂C MXene

National Institute Of Technology Karnataka

Surathkal, Karnataka

- Performed Quantum ESPRESSO-based DFT calculations for Nb₂C MXene
- Calculated **band structure, density of states (DOS), and projected density of states (PDOS)**

TECHNICAL SKILLS

- Good skills in data analysis software: Origin, X'Pert Highscore, QualX, FullProf, EC-Lab, EIS Spectrum Analyzer.
- Basic Python; proficient in MS Office; hands-on experience with XRD, UV-Visible, EIS spectroscopy, IV, Conductivity measurements.

STRENGTH

- Strong theoretical and practical foundation in condensed matter physics and material science
- Quick learner, research-oriented, and highly motivated
- Good communication and teamwork abilities

PROJECT INTEREST

- Design and fabrication of perovskite and thin-film solar cells
- Optical and electrical characterization of photovoltaic materials
- Energy band gap tuning and stability improvement

LANGUAGES

- English - Proficient
- Hindi - Native

HOBBIES

- Reading books
- Watching science-fiction movies